American University of Beirut
EECE 502 – Final Year Project
Course Coordinator – Dr. Youssef Tawk
Minutes for the Weekly Group Meeting – Submitted per Group

Meeting #	Date:	Time:	Duration:	Meeting called by:	Minutes Taker:
2	1/28/26	9 am	1	◉ Advisor ○ Students	Raghad Al Agha

Attendees:

Raghad Al Agha, Lynn Ariss, Hadi Elham, Claritta Khoury

Briefly summarize the main discussions during the meeting:

The team worked on preparing the dataset and training the model, including strategies for dividing the data into 5-seconds chunks and optimizing the size of the model. The team also created a preliminary model using a pipeline-based framework to better understand system behavior and improve performance. The team also worked on analyzing the dataset, which consisted of 105 subjects, and discussed clinical relevance in terms of different features. In terms of hardware, issues such as sensor reading inconsistencies were noted when it comes to the velostat. The team also established a clinical collaboration with Dr. Samia Khoury and worked on preparing the IRB application, which will include a pilot study of 5-10 subjects. The team also emphasized patient comfort and simplicity in terms of testing protocols, with a focus on wearable devices that are simple to use. The team also worked on sourcing locally and manufacturing locally, with some challenges but an optimistic outlook in terms of flexibility in design.

Briefly summarize the conclusions drawn regarding the above-mentioned discussions:

The team agreed to continue optimizing both the model and data processing methods, focusing on achieving a balance between performance and scalability. We concluded that expanding the sensor coverage by using multiple layers is necessary to address hardware issues. We also discussed material selection in order to fit the IP ratings for biomedical wearables. Clinical collaboration with Dr. Khoury is moving forward, and the IRB application is being prepared for a pilot study. The team reiterated the importance of a patient-centered design approach, ensuring that wearable devices and testing protocols are simple, comfortable, and accessible. They also decided to explore local manufacturing and sourcing options to overcome resource constraints, with confidence in the iterative testing and refinement of their design.

Enumerate the assigned tasks by the FYP advisor for each student + the deadline for delivery:

Assigned Task / Per Student	Name of the Student	Deadline
Figure out charging technique of wearable, ordering missing components, trying to generate a heatmap from velostat, research how to change IMU data to steps, finish CITI Course, translate IRB Consent Form	Raghad Al Agha	Wednesday, Feb.4,2026
Work on AI model integration, prepare the flyer for the IRB application, and develop the algorithm for detecting peaks in the PPG sensor data, finish CITI Course	Lynn Ariss	Wednesday, Feb.4,2026
Achieve communication between the two arduinos, finish CITI Course, work on the AI model for ataxia	Hadi Elham	Wednesday, Feb.4,2026
Complete the IRB application, finalize the IRB proposal and IRB consent form and work on presentation.	Claritta Khoury	Wednesday, Feb.4,2026